

Case Study

AI-based Dropout Prediction + Anonymous Counseling — India context

1) Background & why anonymous channel chahiye

- Research aur pilots dikhate hain ki **students mental-health / dropout issues me stigma aur privacy concerns ke karan help lene se khatre** hain; isliye *anonymous, text-based* channels ko zyadatar students prefer karte hain. [PMC+1](#)

2) India examples / pilots (kya already hua hai)

- **AIIMS “Never Alone”** — AIIMS ne students ke liye AI-powered web/WhatsApp accessible counselling platform launch kiya — purpose: 24×7 screening, anonymous initial support, aur referral to human counselors when needed. Iska aim stigma kam karna aur easy access provide karna hai. [The Times of India+1](#)
- **Vidyaarthi Nestham / Govt pilots** — state-level early-warning systems (EWS) use kiye gaye jahan school data + teacher inputs se at-risk students identify karke targeted follow-up kiya gaya (examples: Andhra Pradesh / Bihar pilots). Yeh EWS teacher dashboard + voice/SMS communication use karte hain. [IndiaAI+1](#)

3) Research on dropout prediction (ML side)

- Multiple studies show ML models (RandomForest, CatBoost, XGBoost, neural nets) student logs/grades/attendance se **early dropout prediction** kar sakte hain; Moodle/activity logs etc. se achha signal milta hai. Iska use EWS me kiya ja sakta hai. [Nature+1](#)

4) Key design insights from these cases + papers (actionable)

1. Two-track system — Predict + Anonymous Support

- *Predict* (EWS): model flags risk using attendance, marks, LMS activity, socio-demographics.
- *Support*: anonymous chatbot / WhatsApp interface + content library + peer groups. (AIIMS model shows WhatsApp/web access works well in Indian campuses). [Nature+1](#)

2. Privacy first (anonymity)

- Studies show users disclose sensitive info more to chatbots if anonymity guaranteed — text preferred over voice/video. Design must avoid forced identity capture at first contact. [PMC](#)

3. Low-friction access channels

- Use channels already used by students: **WhatsApp, SMS, web/college portal** — proven effective in Indian pilots (Vidyarthi Nestham used mobile app/teacher workflows). [IndiaAI](#)

4. Escalation rules + human-in-loop

- Bot handles screening & basic counselling; **only** on high-risk flags or consent, human counselor notified with necessary (but minimal) context. This reduces mentor overload and respects student comfort. (Seen in AIIMS concept & chatbot studies). [The Times of India+1](#)

5. Localized, culturally-relevant content

- Region languages, culturally sensitive prompts, short micro-interventions (videos/text) increase engagement. [ResearchGate](#)

5) Practical solution blueprint you can propose (India-ready)

A) Prediction (Backend)

- Data: attendance, marks, LMS logs, last-month absence, discipline records, scholarship/financial flag, distance from college.
- Model: train ensemble (CatBoost / RandomForest) for probability score; threshold for Low/Medium/High. (Train/test on historical data or synthetic if

needed). Nature

B) Anonymous Support Layer (Frontend / Student touchpoint)

- Provide **multi-channel anonymous entry points**:
 - WhatsApp chatbot (preferred in India) — students message a number, remain anonymous.
 - Web chat widget (college portal) — optional login-less access via one-time token.
- Chatbot features:
 - Quick mood check (1–3 Qs), empathetic dialogue, tailored micro-content (videos/articles), self-help exercises.
 - Option to join peer group chats (moderated) or schedule anonymous voice/text session.
- Escalation: if chatbot detects suicidal ideation or very high risk → notify counselor with anonymous token + suggested context. The Times of India+1

C) Mentor Dashboard

- Shows flagged students (by ID) with risk score and minimal redacted context; allows counselor to send an opt-in anonymous message (via bot) asking if student wants human contact — *student chooses to reveal or not*. This reduces forced outreach.

D) Privacy / Compliance

- Store minimal PII, encrypt data at rest, keep opt-in logs, follow local guidelines (Indian IT Act / institutional policies). Ensure counselors trained in ethics.

6) Evidence of effectiveness & lessons

- Pilots (Bihar/Andhra) with EWS + teacher interventions brought back many kids by enabling targeted outreach (shows that prediction + low-cost follow up works). Creative Associates International+1
- AllMS “Never Alone” shows Indian institutions now adopting **AI + WhatsApp/web** approach to reduce stigma and increase access. The Times of India